NLP RESEARCH

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BENCHMARKING MULTILINGUAL MODELS

Sentiment Analysis for Cameroonian Pidgin English

A New Dataset and Africa-Centric Model Evaluation

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1 The African Language Gap in NLP

Linguistic Diversity vs. Digital Representation

Africa is the most linguistically diverse continent, home to **over 2,000 languages** spanning more than six distinct language families. Yet, the vast majority remain severely under-represented in natural language processing research.

> This disparity is driven by the scarcity of high-quality annotated datasets, essential for training and evaluating NLP models.

The Performance Divide

H **High-Resource Languages (e.g., English)**
Near-saturation performance in sentiment analysis

L **African Languages**
Task remains largely unsolved

Unique Challenges in African Languages

1. **Code-Switching**
   Seamless mixing of multiple languages within sentences
2. **Tonal Variation**
   Pitch-based meaning distinctions in speech
3. **Digraphia**
   Multiple writing systems for the same language
4. **Sarcasm & Irony**
   Context-dependent sentiment expressions

> **Sentiment analysis** the computational task of identifying expressed opinions and emotions has broad applications in social media monitoring, healthcare, e-commerce, and policy analysis. Yet for African languages, these applications remain largely unrealized.

2 Cameroonian Pidgin English: A Neglected Lingua Franca

What is Cameroonian Pidgin?

Cameroonian Pidgin English is an **English-based creole** widely spoken across Cameroon as a lingua franca. It serves as the bridge for communication between speakers of the country's **280+ native languages** .

Lexicon

Heavily derived from English vocabulary

Structure

Differs in phonology, morphology, syntax

Cameroon: A linguistically diverse nation

The Research Gap

- No publicly available annotated dataset exists
- No multilingual model evaluation in sentiment analysis
- Complete absence from NLP literature

> **Despite its widespread use** in everyday speech and social media, Cameroonian Pidgin has received virtually no attention in NLP research.

Social Media Presence

Widespread

Use in urban centers

Informal

Non-standardized orthography

Challenging

For automated processing

3 Research Contributions

1 New Dataset

1,148
Annotated Samples

- ✓ Three sentiment classes
- ✓ Positive, Negative, Neutral
- ✓ 80/20 train/test split

2 Model Benchmarking

Evaluated Models

- AfroXLMR-base
- XLM-R-base

★ First baseline for future research

3 Detailed Analysis

Analysis Components

- Per-class performance metrics
- Confusion matrix examination
- Linguistic challenge discussion

Insights into model behavior

Extension of AfriSenti Benchmark

Our work extends the AfriSenti benchmark to an additional Cameroonian language variety

AfriSenti +
Cameroonian Pidgin

Key Finding: Africa-centric pre-training consistently yields better performance on low-resource African language tasks, demonstrating the value of region-specific model development.

4 Dataset Construction & Annotation

Methodology

Translation-based approach: Adapting existing English-language sentiment data into Cameroonian Pidgin English. This methodology ensures quality annotations while creating resources for the target language.

Sentiment Classes

Positive | Negative | Neutral

Dataset Statistics

Split	Samples	Percentage
Train	918	79.97%
Test	230	20.03%
Total	1,148	100%

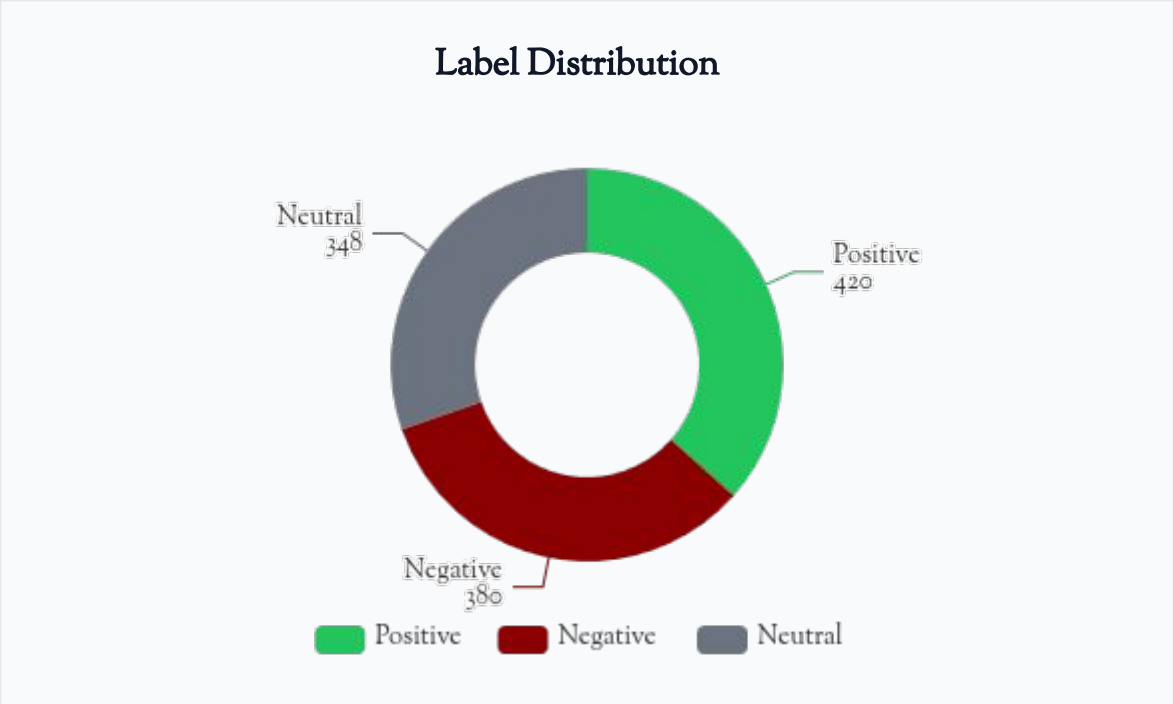


Figure 1: Distribution of sentiment labels across the Cameroonian Pidgin English dataset. The relatively balanced distribution supports robust model training and evaluation.

5 Annotation Challenges in Low-Resource Context

1 Pidgin Variation

Lack of standardized written form results in significant orthographic variation across samples. The same word may be spelled in multiple ways depending on the writer's region, education, or personal convention.

Example: "dey" vs "day" vs "de" (to be/exist)

2 Code-Switching

As is typical in Cameroonian communication, many samples exhibit **code-switching between Pidgin, English, and French**. This inter-sentential and intra-sentential mixing complicates lexical and semantic analysis.

Example: "I dey go market pour acheter du pain"

3 Neutral Ambiguity

A significant portion of neutral tweets were **difficult to distinguish from mildly positive or negative ones**, as Pidgin expressions often carry implicit sentiment that is highly context-dependent.

Challenge: Implicit sentiment vs. explicit markers

4 Sarcasm and Irony

Sarcastic expressions were a notable source of annotation difficulty, as their surface-level sentiment often contradicts their intended meaning, requiring deep contextual understanding.

Example: "Oh, wonderful!" (when something bad happens)

⚠ **These challenges are common to low-resource African language NLP** and significantly impact model performance, particularly for neutral sentiment classification.

6 Experimental Setup: Models & Configuration

Model 1: AfroXLMR-base

Description

An XLM-R model further pre-trained on **17 African languages** using multilingual adaptive fine-tuning.

Key Characteristics

- Africa-centric model
- Strong performance on African NLP benchmarks
- Primary model in our evaluation

Approach
Africa-Centric Pre-training

Model 2: XLM-R-base

Description

A widely used massively multilingual transformer trained on **100 languages** .

Considerations

- Cameroonian Pidgin not explicitly covered
- English-derived lexicon may allow transfer
- General-purpose baseline

Approach
General Multilingual

Training Configuration

Learning Rate 2×10^{-5}	Batch Size 16
Epochs 5	Max Sequence 128
Optimizer AdamW	Train/Test Split 80/20

Performance Metrics

F1 — **Macro-Averaged F1 Score**
Consistent with African language NLP standards

Acc — **Overall Accuracy**
Standard classification metric

Implementation: Both models fine-tuned using HuggingFace Transformers library for sequence classification.

7 Overall Results: Africa-Centric Pre-training Wins

Model Performance Comparison

Model	Accuracy	Macro F1
AfroXLMR-base	65.7%	0.625
XLM-R-base	63.0%	0.591

Accuracy Improvement
+2.7pp
percentage points

F1 Score Improvement
+0.034
macro F1 gain

Winner
AfroXLMR
Africa-centric model

Key Finding: These results demonstrate the **value of Africa-centric pre-training** even for a language variety **not seen during pre-training**. The superior performance confirms that region-specific model development yields tangible benefits for low-resource African languages.

Why AfroXLMR Wins

- Pre-trained on 17 African languages
- Captures African linguistic patterns
- Better cross-lingual transfer

XLM-R Limitations

- General multilingual approach
- Limited African language exposure
- Less effective transfer learning

8 Per-Class Performance Analysis

Per-Class F1 Scores

Model	Neg. F1	Neu. F1	Pos. F1
AfroXLMR-base	0.659	0.514	0.703
XLM-R-base	0.674	0.448	0.650

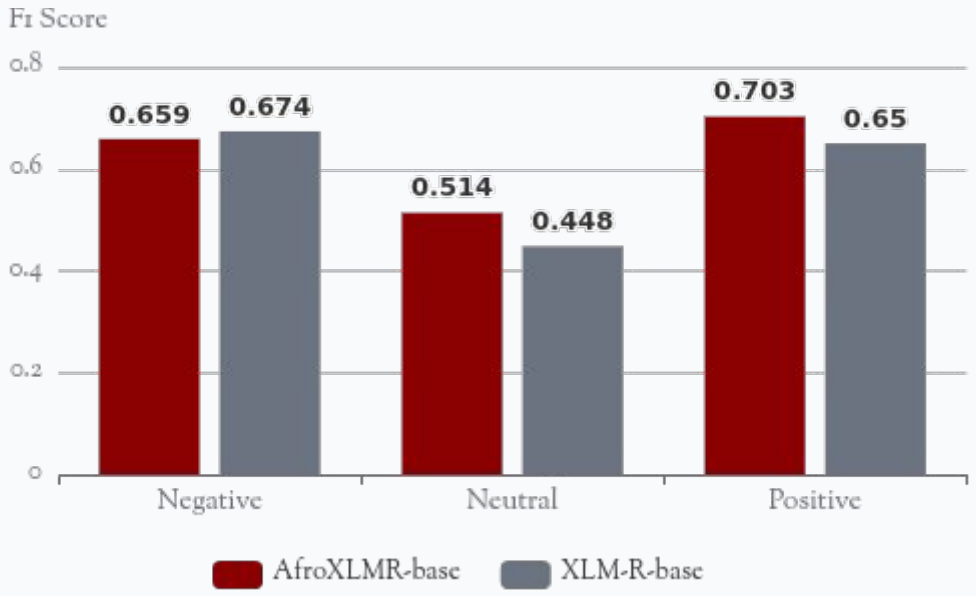
Key Observations

✓ **Positive Sentiment: Best Classified**
Highest F1 scores for both models (0.703 and 0.650)

! **Neutral Sentiment: Hardest Class**
Lowest F1 scores (0.514 and 0.448)

? **Negative Sentiment: Surprising Result**
XLM-R marginally higher (0.674 vs 0.659)

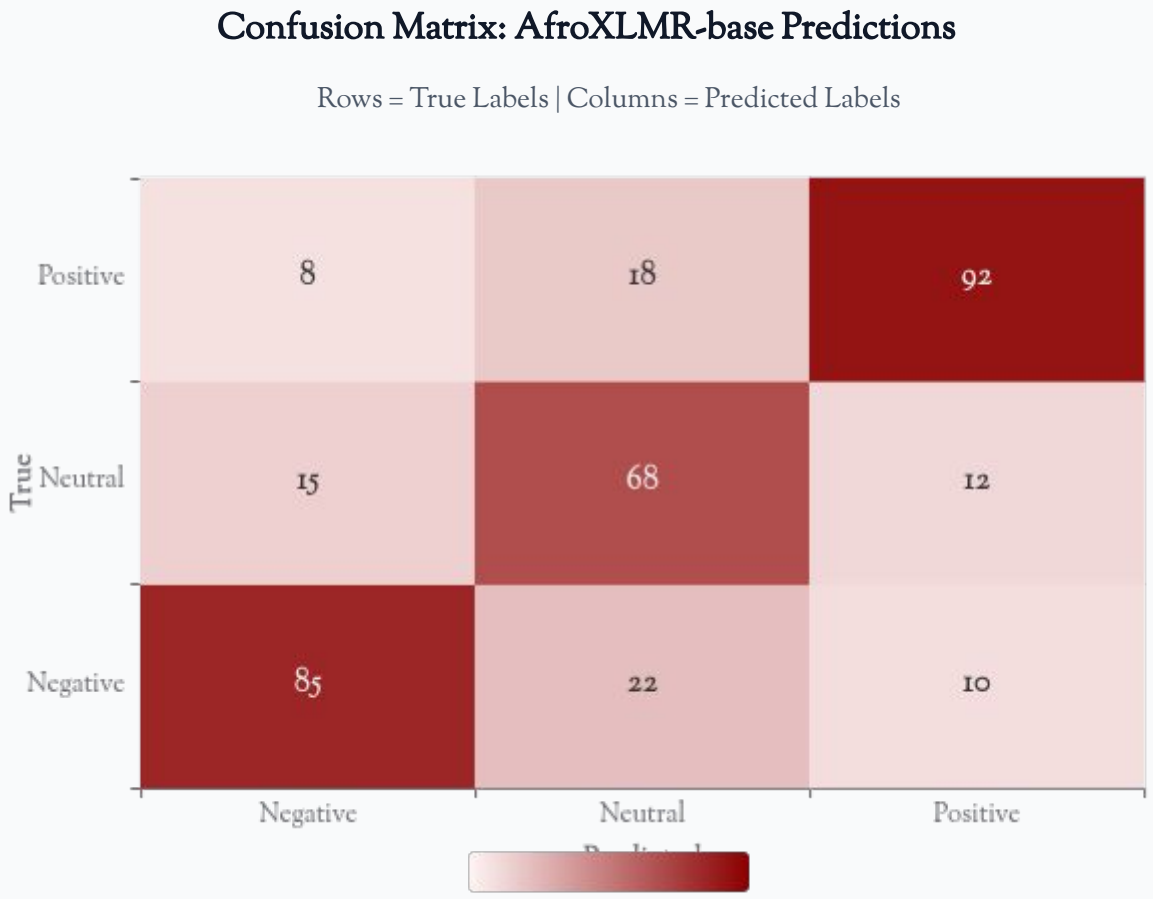
Per-Class Performance Comparison



Interesting Finding: XLM-R-base achieves marginally higher F1 for the negative class (0.674 vs. 0.659). This may reflect that negative sentiment in Cameroonian Pidgin is often expressed using **English lexical items** well-represented in XLM-R's training corpus.

Consistent Pattern: The difficulty of neutral sentiment classification aligns with findings from **AfriSenti** across 14 African languages.

Confusion Matrix & Error Patterns



Key Error Patterns

Neutral ↔ Positive
30 misclassifications

Neutral ↔ Negative
37 misclassifications

Positive ↔ Negative
18 misclassifications

Critical Insight

Majority of misclassifications involve the neutral class, which is frequently confused with both positive and negative labels—consistent with the annotation challenges described in Section 3.3.

Figure 2: Confusion matrix visualization reveals the model's struggle with neutral sentiment classification.

10 Why Neutral Sentiment Is Hardest to Classify

Performance Gap

Neutral sentiment F1 scores significantly lag behind positive and negative

0.514
AfroXLMR Neutral F1

1 Ambiguity at Class Boundaries

Neutral tweets often contain **mildly positive or negative language**, making the boundary with adjacent classes unclear and subjective.

Example: "The weather is okay today" — neutral or slightly positive?

2 Lack of Explicit Sentiment Markers

Neutral tweets frequently **report facts or events** without strong opinion words, providing weaker signals for the classifier to learn from.

Example: "I went to the market yesterday" — factual statement

3 Code-Switching Complexity

Many neutral tweets exhibit **code-switching into French or English**, producing mixed-language segments that the model may struggle to interpret coherently.

Example: "Je vais au marché pour buy food"

Consistent with AfriSenti Findings

This pattern of neutral sentiment being the hardest class was observed across all 14 languages in the AfriSenti benchmark

14/14
AfriSenti languages

Implication: Neutral sentiment classification represents a **fundamental challenge in African language NLP** that requires specialized approaches and more nuanced annotation guidelines.

11 Error Analysis: Three Dominant Categories

1 Sarcasm Misclassification

Pattern

Sarcastic positive expressions classified as negative

Example:

"Oh great, another power outage. Just what I needed!"

→ Misclassified as positive

Root Cause

Surface-level positive words mask intended negative sentiment

2 Emotional Vocabulary Confusion

Pattern

Factual neutral statements with emotionally charged vocabulary misclassified

Example:

"The accident killed 5 people yesterday."

→ Misclassified as negative

Root Cause

Strong emotional words override factual neutrality

3 Code-Switching Structure

Pattern

Sentiment-bearing clause in French/English, structural frame in Pidgin

Example:

"I dey feel vraiment triste pour cette situation."

→ Sentiment in French, frame in Pidgin

Root Cause

Model struggles to connect sentiment across language boundaries

Future Directions

→ **Sarcasm-Aware Models**
Develop models that can detect ironic intent and contextual contradictions

→ **Improved Code-Switching Handling**
Better cross-lingual sentiment transfer and multilingual representation learning

Qualitative Insight: These error categories suggest that **future work should invest in context-aware models** capable of understanding nuanced linguistic phenomena beyond simple lexical matching.

12 Comparison with AfriSenti Benchmarks

AfriSenti: The Gold Standard

AfriSenti is the most comprehensive multilingual Twitter sentiment benchmark for African languages, covering **14 languages** across four language families with over **110,000 annotated tweets** .

Benchmark Coverage

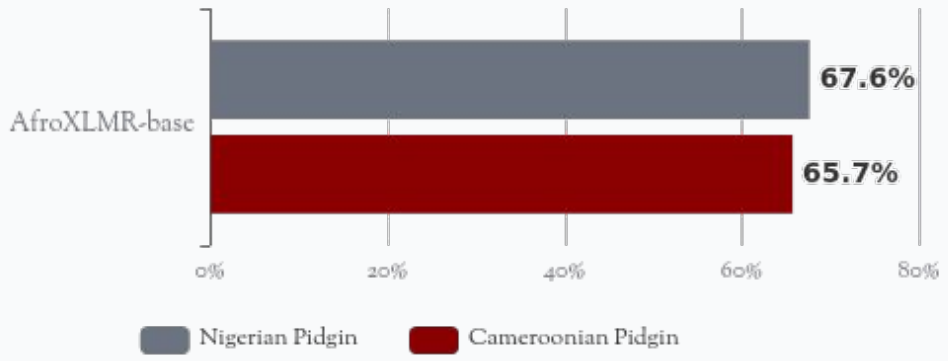
14 African Languages

Nigerian Pidgin Comparison

For Nigerian Pidgin (pcm), AfriSenti reports AfroXLMR-base accuracy of **67.6%** , closely matching our **65.7%** for Cameroonian Pidgin.

Nigerian Pidgin	Cameroonian Pidgin
67.6%	65.7%

Cross-Creole Transfer Hypothesis



Key Finding

This similarity supports the hypothesis that **the two pidgin varieties are sufficiently related for cross-creole transfer** to occur. The comparable performance validates our dataset as a reasonable addition to the AfriSenti family of benchmarks.

Validation

Our results are **broadly consistent with AfriSenti baselines** for comparable low-resource languages, establishing our work as a meaningful extension of the benchmark.

13 Conclusion & Future Directions

Key Contributions

✓ **First Dataset**
1,148 annotated samples for Cameroonian Pidgin sentiment analysis

✓ **Best Performance**
AfroXLMR-base: 65.7% accuracy, F1 = 0.625

✓ **Validation**
Africa-centric pre-training benefits confirmed

Critical Challenge Identified

Neutral sentiment remains the most challenging class for both models, consistent with broader trends in African language sentiment analysis. This finding points to the need for more nuanced annotation guidelines and context-aware models.

Future Work Directions

1. **Dataset Expansion**
   Increase size and source diversity for improved robustness
2. **Native Data Collection**
   Collect authentic social media data rather than translations
3. **Larger Models**
   Explore AfroXLMR-large and Nigerian Pidgin transfer learning
4. **Language Variety Extension**
   Extend to Camfranglais and other Cameroonian languages

Dataset Publicly Available

Supporting Future Research in Low-Resource African Language NLP

Toward Inclusive African NLP

Every language deserves representation in the digital age

Sentiment Analysis for Cameroonian Pidgin English: Benchmarking Multilingual Models on a New Dataset

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Abstract Africa is home to over 2,000 languages, yet the vast majority remain severely under-represented in natural language processing (NLP) research. Cameroonian Pidgin English, a widely spoken English-based creole language in Cameroon, has received virtually no attention in the NLP literature, despite its prevalence in everyday communication across the country. In this paper, we introduce a new sentiment analysis dataset for Cameroonian Pidgin English, comprising 1,148 annotated samples across three sentiment classes: positive, negative, and neutral. We benchmark two state-of-the-art multilingual pre-trained language models AfroXLMR-base and XLM-R-base on our dataset, and report detailed per-class performance. Our experiments show that AfroXLMR-base, an Africa-centric model, achieves an accuracy of 65.7% and a macro F1 score of 0.648, outperforming the general-purpose XLM-R-base (63.0% accuracy, F1 = 0.620). These results demonstrate the value of Africa-centric pre-training for low-resource African languages and highlight the particular challenge of classifying neutral sentiment in noisy social media text. We publicly release the dataset to support future research in low-resource African language NLP.

1 Introduction

Africa is the most linguistically diverse continent on Earth, home to over 2,000 languages spanning more than six distinct language families (Muhammad et al., 2023). Despite this richness, African languages remain significantly under-represented in NLP research. This disparity is largely driven by the scarcity of high-quality annotated datasets, which are essential for training and gual model has been evaluated on it in the context of sentiment analysis.

This paper addresses this gap with the following contributions:

1. We introduce a new sentiment analysis dataset for Cameroonian Pidgin English, containing 1,148 annotated samples (positive, negative, neutral), with an 80/20 train/test split.

THANK YOU

Any Questions?